

NEO Mathematics Learner Guide

Straight-Line Graphs - $y = mx + c$

Can a line tell the story of change?

What are we exploring?

You will explore how a straight line can show a starting point and a rate of change. You will change graphs, match different representations, solve line clues and create a linear model of your own.

Your Stepping Stones

Connect - equations, tables and graphs.

Find - the gradient.

Find - the y-intercept.

Explain - what a line is showing.

Create - a straight-line model.

A possible way through

Start here: Read the Driving Question.

Explore: Use the Gradient and Intercept Explorer.

Predict: Sketch a line on the coordinate Scratchpad before testing it.

Connect: Match a story, equation, table and graph description.

Detect: Find m , c and an equation in Line Detective.

Pause: Use the Rest section whenever you need it.

Create: Build a human story for a line.

Practise: Open the Practice Companion.

You can move backwards and forwards. You do not have to complete every section in one sitting.

When you feel stuck

1. Start: where does the line cross the y-axis?
2. Change: what happens to y when x increases by 1?
3. Check: does the equation match the table, graph or story?
4. Use the coordinate Scratchpad to plot two points or sketch a gradient triangle.
5. Change one thing at a time.

You can also use: **Socratic prompt**, **Hint**, **Check my answer**, **Worked solution** and the **NEO Maths Scratchpad**.

Using support is part of learning. It is not cheating.

What you might need

Useful

- Device
- Paper or notes app

You might also like

- Ruler
- Graph paper
- Calculator
- Two coloured pens - one for the start and one for the change
- The embedded Coordinate Scratchpad

How will I know I am making progress?

You may be making progress when you can say:

"I can find the starting value."

"I can find the rate of change."

"I can connect m with gradient."

"I can connect c with the y -intercept."

"I can explain what a line means in a real situation."

You do not need all of these to be true yet. They are signs of where the learning is heading.

Before you finish

One thing I noticed...

One thing that feels clearer...

One thing I may return to...

The support that helped me most was...

Your Scratchpad

The Scratchpad stays inside the lesson. You can draw, annotate, make a prediction, use a grid or clear the board and begin again. You can use paper instead. The Scratchpad is there to help you think.